Composite Trapezoidal Rule with a Certified Error Bound

An End-to-End Meta-Math Walkthrough

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Abstract

A nonnegative kernel expresses the local quadrature error exactly. Its integral gives the constant, and summing local errors gives the global order. The report presents the target theorem, declared imports, a dependency graph, five-component node contracts, a continuous proof, a worked example, and a hypothesis-repair audit. It is an authored mathematical exposition, not a deployed-agent transcript or a formal proof-assistant certificate.

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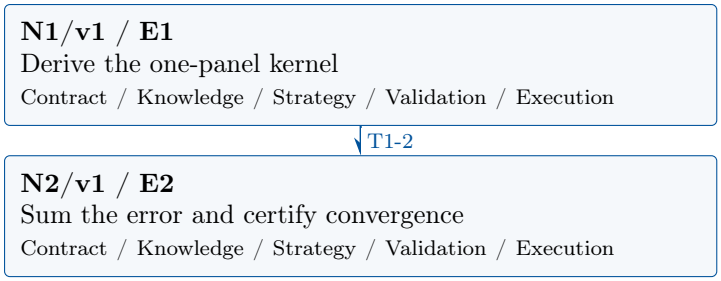
I. TARGET AND IMPORT CONTRACT

Theorem I.1. *For $f \in C^2([a,b])$, $a < b$, let $N \geq 1$, $h = (b-a)/N$, and $x_j = a + jh$. Define $T_N = h(f(a)/2 + \sum_{j=1}^{N-1} f(x_j) + f(b)/2)$. Prove $|T_N - \int_a^b f(x)\,dx| \leq (b-a)h^2\|f''\|_\infty/12$. If $f'' \geq 0$, also prove $T_N \geq \int_a^b f$. These concern exact sampled values and arithmetic.*

A. Foundations and permitted imports

Use integration by parts for C^2 functions and finite sums. The bound $\|f''\|_\infty$ is finite by continuity on the compact interval. Distinguish quadrature error from errors in function evaluation or floating-point summation, which require separate estimates.

II. NODE INTERACTION WORKFLOW



Each box is one mathematical agent containing all five functional components. Blue arrows carry the versioned exports listed below. The layout gives a reading order; only drawn arrows assert dependencies. Diagnostic repair is separate from the proof DAG.

A. Versioned evidence-transfer ledger

T1-2: N1/E1/v1 to N2. $\frac{h}{2}(f(u)+f(v)) - \int_u^v f = \int_u^v K(t) f''(t)\, dt.$

Receiving activity. N2.1

Compatibility. E1/v1 on each of the N equal panels. Check: There are $N = (b-a)/h$ panels, so the local h^3 bound becomes a global h^2 bound.

III. AGENT ROLES AND FIVE-COMPONENT ACTIVITIES

A. N1/v1: Derive the one-panel kernel

Object and role. Integrate Kf'' twice by parts to recover endpoint weights and the exact integral.
Local obligation: $\frac{h}{2}(f(u)+f(v)) - \int_u^v f = \int_u^v K(t)f''(t)\,dt$.

Inputs. One interval $[u,v]$ of length h and $K(t) = (t-u)(v-t)/2$.

C — Contract

Publish E1/v1 only after establishing this obligation: $\frac{h}{2}(f(u)+f(v)) - \int_u^v f = \int_u^v K(t)f''(t)\,dt$.

K — Knowledge

One interval $[u,v]$ of length h and $K(t) = (t-u)(v-t)/2$. Retain the stated assumptions and import versions.

S — Strategy

Integrate Kf'' twice by parts to recover endpoint weights and the exact integral.

V — Validation

The boundary values are $K(u) = K(v) = 0$, $K'(u) = h/2$, $K'(v) = -h/2$, and $K'' = -1$.

E — Execution

Carry out activities N1.1–N1.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N1.1 Integration by parts gives $\int_u^v Kf'' = -\int_u^v K'f'$, since K vanishes at the endpoints. Integrating again gives $-[K'f]_u^v + \int_u^v K''f$.

N1.2 Using the displayed endpoint derivatives and $K'' = -1$ yields $(h/2)(f(u)+f(v)) - \int_u^v f$, exactly the local error with the sign $T - I$. Also $K \geq 0$ and, after setting $s = t-u$, $\int_u^v K = \frac{1}{2}\int_0^h s(h-s)\,ds = h^3/12$.

Export E1/v1. $\frac{h}{2}(f(u)+f(v)) - \int_u^v f = \int_u^v K(t)f''(t)\,dt$.

Acceptance gate. The boundary values are $K(u) = K(v) = 0$, $K'(u) = h/2$, $K'(v) = -h/2$, and $K'' = -1$.

Targeted repair. If the sign of the error is uncertain, recompute the endpoint derivatives of K before taking absolute values. Withhold this export and recheck its descendants after a versioned correction.

B. N2/v1: Sum the error and certify convergence

Object and role. Add exact identities, then use nonnegativity of the kernel for magnitude and sign.
Local obligation: The stated absolute error and convex-function overestimate hold; in particular $T_N \to \int_a^b f$.

Inputs. E1/v1 on each of the N equal panels. Incoming proof exports: E1/v1 at N2.1.

C — Contract

Publish E2/v1 only after establishing this obligation: The stated absolute error and convex-function overestimate hold; in particular $T_N \to \int_a^b f$.

K — Knowledge

E1/v1 on each of the N equal panels. Incoming proof exports: E1/v1 at N2.1. Retain the stated assumptions and import versions.

S — Strategy

Add exact identities, then use nonnegativity of the kernel for magnitude and sign.

V — Validation

There are $N = (b-a)/h$ panels, so the local h^3 bound becomes a global h^2 bound.

E — Execution

Carry out activities N2.1–N2.2 below, retaining each calculation and construction as evidence.

Numbered mathematical activities

N2.1 Summing the one-panel trapezoids counts each interior sample twice with weight $h/2$, producing T_N. The exact error is the sum of the panel kernel integrals. Its absolute value is bounded by $N(h^3/12)\|f''\|_\infty = (b-a)h^2\|f''\|_\infty/12$.

N2.2 If $f'' \geq 0$, every kernel integral is nonnegative, proving the overestimate. Since $h \to 0$ as $N \to \infty$, the absolute bound also proves convergence. The smoothness and fixed interval make the same derivative bound valid for all N.

Export E2/v1. The stated absolute error and convex-function overestimate hold; in particular $T_N \to \int_a^b f$.

Acceptance gate. There are $N = (b-a)/h$ panels, so the local h^3 bound becomes a global h^2 bound.

Targeted repair. If the local h^3 rate is reported as the global rate, multiply by the number of panels and restore the h^2 bound. Withhold this export and recheck its descendants after a versioned correction.

IV. CONTINUOUS MATHEMATICAL PROOF

A. N1 — Derive the one-panel kernel

1. **N1.1.** Integration by parts gives $\int_u^v Kf'' = -\int_u^v K'f'$, since K vanishes at the endpoints. Integrating again gives $-[K'f]_u^v + \int_u^v K''f$.

2. **N1.2.** Using the displayed endpoint derivatives and $K'' = -1$ yields $(h/2)(f(u) + f(v)) - \int_u^v f$, exactly the local error with the sign $T - I$. Also $K \geq 0$ and, after setting $s = t - u$, $\int_u^v K = \frac{1}{2}\int_0^h s(h-s)\,ds = h^3/12$.

B. N2 — Sum the error and certify convergence

1. **N2.1.** Summing the one-panel trapezoids counts each interior sample twice with weight $h/2$, producing T_N. The exact error is the sum of the panel kernel integrals. Its absolute value is bounded by $N(h^3/12)\|f''\|_\infty = (b-a)h^2\|f''\|_\infty/12$.

2. **N2.2.** If $f'' \geq 0$, every kernel integral is nonnegative, proving the overestimate. Since $h \to 0$ as $N \to \infty$, the absolute bound also proves convergence. The smoothness and fixed interval make the same derivative bound valid for all N.

□

V. WORKED ORCHESTRATION

For $f(x) = x^2$ on $[0,1]$ and $N = 2$, $T_2 = \frac{1}{2}(0 + \frac{1}{4} + \frac{1}{2}) = 3/8$, while the integral is $1/3$. The error is $1/24$, exactly the bound $1 \cdot (1/2)^2 \cdot 2/12 = 1/24$. Convexity correctly predicts an overestimate.

A. Boundary and hypothesis audit

For $f(x) = |x|$ on $[-1,1]$, the one-panel trapezoid is 2 and the integral is 1. Although $f'' = 0$ away from zero, the function is not C^2; treating its a.e. second derivative as a classical uniform bound would incorrectly predict zero error.

VI. ILLUSTRATIVE FAILURE AND REPAIR

The following is an authored diagnostic scenario. Changed assumptions define a new contract; they do not retroactively prove the original statement.

A. Rejected argument, diagnosis, and correction

Rejected N1 shortcut: use $f'' = 0$ almost everywhere in the C^2 kernel formula for $|x|$. **Diagnosis:** Knowledge silently replaced classical regularity by an a.e. derivative without the needed absolute-continuity bridge. **Repair:** keep the stated C^2 hypothesis; for $|x|$, split at its corner and integrate the two affine pieces separately, obtaining the exact value 1. That split is a separate valid computation, not a proof of the unqualified bound.

B. Publication and reuse

The corrected export retains its explicit assumptions. A hypothetical revision receives a new version, and each consuming activity is rechecked; this scenario is not a recorded runtime event. **Next direction:** Develop adaptive panel selection from local derivative bounds, then add explicit evaluation and roundoff error terms for numerical certification.

Provenance. This authored proof is complete relative to its declared imports. Reuse each export with its assumptions, proof, and version. The original Markdown remains available for comparison.

Further reading. Boyd and Vandenberghe, Convex Optimization for convexity, optimality, and descent; University of Maryland, Scientific Computing I for the numerical-analysis setting. The estimates used here are derived explicitly in §4.